Towards a Next-Generation Agricultural Expert System

A Multimodal Conversational Approach

A report submitted in partial fulfilment of the requirements for the award of degree of

M.Tech.

in

Computer Science (Artificial Intelligence)

by

Shajid Ali Khan

(MCS25049)

under the guidence of

Dr. Deepak Kumar Singh



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY LUCKNOW-226002

Jul-Sep, 2024

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ACKNOWLEDGEMENT

I am highly indebted to **Supervisor Name**, and obliged for giving me the autonomy of functioning and experimenting with ideas. I would like to take this opportunity to express my profound gratitude to him not only for his academic guidance but also for his personal interest in my report and constant support coupled with confidence boosting and motivating sessions which proved very fruitful and were instrumental in infusing self-assurance and trust within me. The nurturing and blossoming of the present work is mainly due to his valuable guidance, suggestions, astute judgement, constructive criticism and an eye for perfection. My mentor always answered myriad of my doubts with smiling graciousness and prodigious patience, never letting me feel that I am novices by always lending an ear to my views, appreciating and improving them and by giving me a free hand in my report. It's only because of his overwhelming interest and helpful attitude, the present work has attained the stage it has.

Finally, I am grateful to our Institution and colleagues whose constant encouragement served to renew my spirit, refocus my attention and energy and helped me in carrying out this work.

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ABSTRACT

Indian agriculture, which supports the livelihoods of over 600 million people, faces a severe advisory deficit. With only one agricultural extension worker for every 1,000+ farmers, crop disease, nutrient deficiency, poor crop selection, and market price asymmetry collectively drive yield losses exceeding 1.5 lakh crore annually. Existing agricultural AI systems — while individually promising — remain fundamentally fragmented: the best deployed chatbot lacks image-grounded disease diagnosis, the best vision model has no conversational interface, and none deliver explainable, multilingual advisory in a single session.

This research proposes, designs, and evaluates a next-generation multi-agent, multi-model agricultural expert system that integrates five specialist AI agents — disease diagnosis, nutrition deficiency, crop recommendation, weather and irrigation, and live market price advisory — orchestrated by an intent-routing backbone built on LangGraph. The system accepts voice, image, text, and soil sensor inputs; grounds all responses in a 250,000+ document RAG knowledge base built from ICAR, FAO, and CGIAR sources; renders farmer-facing Grad-CAM heatmaps and SHAP explainability charts; and delivers advice in ten Indic languages via Whisper ASR and Bhashini TTS.

Experimental evaluation targets greater than 90% top-1 accuracy on disease classification across 200+ diseases and 50+ crops, RAG faithfulness above 0.88, word error rate below 20% for Hindi agricultural speech, and a mean agronomist agreement score above 4.0/5.0 from a panel of ten ICAR-certified experts. The system is designed for deployment as a Progressive Web Application accessible on commodity Android devices with 3G connectivity. This work makes five original contributions to the agricultural AI literature and directly closes five specific gaps unaddressed by the current state of the art.

Keywords: Multi-agent systems · Vision-language models · Retrieval-augmented generation · Explainable AI · Indic NLP · Agricultural advisory · Precision agriculture · Multimodal AI

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1 Introduction

This chapter establishes the motivation, context, and framing for the proposed research. It presents the scale of India's agricultural advisory deficit, examines the current state of AI-driven agricultural systems, identifies why those systems remain insufficient, and articulates the research questions that drive this thesis. Supporting statistics, comparative figures, and structured tables are provided throughout to ground each claim in evidence.

1.1 Background and Context

Agriculture is the structural backbone of the Indian economy. It contributes approximately 17–18% of GDP, employs 42.3% of the total workforce, and directly supports the livelihoods of over 600 million people across 146 million farm households [15]. Despite this scale, Indian agriculture continues to operate under a profound and well-documented advisory deficit — a gap between the technical knowledge needed to make optimal decisions and the access farmers have to that knowledge.

The magnitude of this gap is quantified starkly in the extension worker ratio. National data from ICAR [16] places India's agricultural extension worker to farmer ratio at 1:1,000+ — far below the FAO-recommended benchmark of 1:400. In densely farmed states such as Uttar Pradesh, Bihar, and Madhya Pradesh, this ratio deteriorates to 1:2,000 or worse. The practical consequence is that the majority of India's smallholder farm households make critical decisions — about disease treatment, fertiliser application, crop selection, irrigation scheduling, and market timing — without any professional guidance.

The economic cost of this advisory gap is not abstract. Four distinct, quantified channels of loss have been identified across the literature and government reports:

- **Crop disease and pest damage (26–30% yield loss):** Farmers identify disease symptoms visually, typically 10–14 days after the optimal treatment window. Misdiagnosis — treating fungal blight as bacterial — leads to wrong pesticide application, wasted money, and crop failure. Only 12% of farmers have used any digital disease tool [17].

- **Soil nutrient deficiency (20–30% yield loss):** Nitrogen, zinc, and boron imbalances silently reduce potential yield by 20–30% annually [17]. The Soil Health Card scheme has issued 230+ million cards, but only 14.6% of recipients acted on the recommendations — because no system translates NPK numbers into a fertiliser schedule in their language.

- **Market price asymmetry (30–40% below-optimal revenue):** Farmers sell an estimated 30–40% below the price achievable with real-time mandi data and price forecasting [18]. Only 22 million of 146 million farm households use eNAM or any digital market platform. The rest sell at distress timing without any price intelligence.

- **Suboptimal crop selection:** Without soil-matched, climate-matched, and historically yield-validated crop recommendations, many farmers plant suboptimal varieties for their conditions. District-level crop diversification data shows that over 60% of smallholders in any given district cultivate fewer than three crop types — a risk concentration driven partly by lack of recommendation access.

Together these four channels account for the estimated 1.5 lakh crore in annual agricultural losses cited by industry reports and corroborated by ICAR internal estimates. This figure represents both an economic tragedy and a clear research opportunity — one that advances in AI are now technically positioned to address.

1.2 The AI Opportunity — Why the Time Is Now

The convergence of several distinct AI technology maturations in 2023–2025 makes the construction of a comprehensive agricultural advisory system both necessary and — for the first time — feasible. Each technology listed below has crossed a threshold of practical utility within the last two years.

Table 1: Technology readiness assessment — components of the proposed system

Technology	**Current Capability**	**Agricultural Relevance**	**Evidence**
GPT-5.5 / Gemini 3.1 Pro	Strong domain reasoning, multi-turn dialogue, native tool use, reduced hallucination vs. prior generation, 1M token context	Orchestrator backbone; advisory response generation and multi-agent coordination	OpenAI (Apr 2026); Google DeepMind (2026)
ViT-L/16 + LoRA fine-tuning	>90% accuracy on agricultural disease datasets; >99% reduction in trainable parameters vs. full fine-tuning	Disease diagnostic agent with expert-level accuracy at minimal GPU cost	[4]; [7]
Whisper large-v3 (Indic fine-tuned)	WER 16.2% for Hindi agricultural speech on real farm recordings; WER 35.1% for Odia	End-to-end voice advisory input in local language; domain fine-tuning required for agricultural vocabulary	[14]

Continued on next page

Table 1 – continued from previous page

Technology	Current Capability	Agricultural Relevance	Evidence
Bhashini API	Production-grade ASR + TTS for 10+ Indic languages; government-backed; free via API	Voice input and spoken response delivery for farmers in Hindi, Tamil, Telugu, Marathi, Kannada, Bengali, Gujarati, Punjabi, Odia, Assamese	MeitY / Bhashini (2024)
Hybrid RAG + Qdrant	Faithfulness >0.88 in domain Q&A; 34 pp improvement over unaugmented LLM baseline; hallucination <5% with LLM-as-judge gating	Hallucination-resistant agronomic knowledge retrieval grounded in ICAR, FAO, and CGIAR verified sources	RAGAS (2024); [19]
LangGraph v1.1 (Dec 2025)	Stateful parallel agent orchestration; graph-based cyclic workflows; persistent checkpointing; production middleware (retry, content moderation); 30k+ GitHub stars	Specialist model routing per query type; multi-turn session state across five agricultural agents	LangChain State of AI Agents Report (2025); [20]
React 18 PWA + FastAPI WebSocket	Offline cache via Workbox service worker; <4 s streaming latency on 3G; SSE token-by-token streaming	Low-connectivity web access for farmers on commodity Android devices; inline Grad-CAM and SHAP rendering	Workbox (2025); FastAPI (2025)

Critically, no prior work has assembled these components into a unified, deployment-ready system. The proposed research addresses this specific integration gap, not individual component research — the novelty lies in the architecture, fusion, and deployment rather than any single model advance.

1.3 Current State of Agricultural AI Systems

A wave of agricultural AI systems has emerged in 2023–2025, including Farmer.Chat, AgriDoctor, AgriGPT, AgriM-LLM, AgroAskAI, and the MDPI Agentic framework. While each system represents a significant step forward in specific areas, none provides a unified, multi-agent, and cross-modal solution tailored to the Indian smallholder

farmer. A detailed review of these architectures, including their specific contributions and limitations, is presented in Chapter 2.

1.4 Problem Statement

The formal problem addressed by this research can be stated as follows:

Formal Problem Statement:
Current agricultural AI systems are fundamentally fragmented and individually insufficient. Farmer.Chat [1] achieves conversational scale but lacks image-grounded disease diagnosis, soil sensor integration, and explainability. AgriDoctor [2] introduces intent-routing multi-agent architecture but is confined to disease and pest identification only. AgriGPT [3] achieves strong knowledge grounding through Tri-RAG but provides no voice input, visual diagnosis, or farmer-facing explanation. AgriM-LLM [4] advances vision encoding for disease but operates as a standalone model with no conversational interface or multilingual support. AgroAskAI [5] demonstrates multi-agent climate advisory with hallucination governance but omits visual diagnosis, soil data, and market intelligence. No existing system simultaneously supports multimodal input across voice, image, text, and soil sensor; routes queries to specialist domain models; grounds responses in verified agronomic knowledge; surfaces explainability in a farmer-accessible form; delivers advisory in Indic languages; or integrates live market price within the same conversational session as agronomic advice — leaving India's 146 million smallholder farmers without a reliable, trustworthy, and accessible AI-driven advisory system.

Figure 1: Formal problem statement for this research.

1.5 Research Questions

The following five research questions are derived directly from the five capability gaps identified in the comparative analysis:

Table 2: Five research questions derived from the five identified capability gaps

RQ	Research Question	Gap Addressed
RQ1	Can a multi-agent architecture with five specialist domain models outperform a single general-purpose LLM for multi-domain agricultural advisory in terms of accuracy, relevance, and latency?	G1 — No unified advisory session
RQ2	Does cross-modal fusion of leaf image features with real-time soil sensor data improve nutrition deficiency diagnosis and fertiliser recommendation accuracy over image-only or sensor-only approaches?	G2 — No sensor-image fusion in dialogue
RQ3	Does rendering Grad-CAM heatmaps and SHAP feature charts to farmers in the advisory response improve trust, acceptance, and correctness verification compared to advice without explanations?	G3 — XAI never shown to farmer
RQ4	Can an end-to-end Indic voice pipeline — Whisper ASR to RAG advisory to Bhashini TTS — achieve production-grade accuracy (WER <20% Hindi) and sub-4-second latency within a deployed agricultural advisory system?	G4 — No Indic voice-to-advisory pipeline
RQ5	Does integrating live mandi price forecasting within the same conversational session as agronomic advisory improve the quality and actionability of farmer decisions compared to advisory systems without market integration?	G5 — Market never co-delivered with agronomy

1.6 Scope and Delimitations

1.6.1 Scope of the Research

- **Geographic scope:** India-focused multi-crop advisory covering 22 states and 50+ crop species across Kharif, Rabi, and Zaid seasons.
- **Input modalities:** Text, voice (10 Indic languages), leaf/field images, soil sensor data (NPK, pH, EC, moisture), and live market price feeds.

- **Advisory domains:** Disease diagnosis, nutrition deficiency and fertiliser recommendation, crop selection and sowing calendar, weather and irrigation scheduling, and market price advisory.
- **Deployment platform:** Progressive Web Application (PWA) targeting commodity Android devices on 3G connectivity with <4 second response latency.
- **Knowledge grounding:** ICAR Package of Practices, FAO advisory documents, CGIAR research corpus, Kisan Call Centre transcripts, and Agmarknet market data.

1.6.2 Delimitations

- This research does not develop new base models — it fine-tunes and integrates existing SOTA architectures (ViT-L/16, Whisper large-v3, GPT-4o).
- IoT sensor hardware design and manufacturing is outside scope — the system assumes sensor data arrives via standard MQTT or REST API.
- The system covers 50+ crops but does not claim exhaustive coverage of all 250+ crops cultivated in India — priority is given to the 10 highest-acreage crops by state.
- Market forecasting covers mandi price prediction — commodities futures and derivatives markets are out of scope.

1.7 Significance of the Research

This research is significant along three independent dimensions:

1.7.1 Scientific Significance

- First multi-agent architecture with five specialist AI domains in one agricultural system
- First cross-modal fusion of leaf image + soil sensor for nutrition advisory in a conversation
- First system rendering Grad-CAM + SHAP as part of a farmer-facing advisory response
- Largest Indian agricultural knowledge corpus for RAG (250k+ ICAR/FAO/CGIAR chunks)
- First end-to-end Indic voice pipeline connected to agronomic RAG advisory

1.7.2 Societal and Economic Significance

- Potential to reach 146M farm households with zero-cost, 24/7 expert advisory
- Closes the Soil Health Card action gap — translates data into language-delivered advice
- Market price integration can recover 30–40% of below-optimal revenue for farmers
- Indic voice pipeline removes literacy and connectivity barriers to AI access
- Explainability increases trust and adoption — critical for marginalised user populations

1.7.3 Policy and Institutional Alignment

- Direct support for three Government of India flagship initiatives: the Soil Health Card Scheme (DAHD, 2015), the National Agriculture Market (eNAM, 2016), and the Digital Agriculture Mission (MoAFW, 2023).
- Grounding in ICAR Package of Practices to ensure alignment with the national agricultural extension framework.
- Integration of the Bhashini API to align with the MeitY Digital Language Inclusion mandate.

2 Literature Review

This chapter presents a comprehensive review of prior research across six streams that form the intellectual foundation of this thesis. For each stream, key papers are analysed in terms of their architectural contributions, empirical results, and — critically — their specific limitations that motivate the proposed research.

2.1 Evolution of Agricultural Expert Systems (1980–2024)

Agricultural expert systems have a history spanning four decades. Understanding this evolution is essential for positioning the proposed research correctly — not as an isolated technical contribution but as the culmination of a long progression from rule-based symbolic systems to neural, conversational, and agentic architectures.

A consistent pattern emerges across this evolution: each generation solves the primary limitation of the previous one but introduces new gaps. Deep learning solved rule brittleness but lost interpretability. LLMs solved knowledge breadth but introduced hallucination. RAG solved hallucination but remained text-only. Multimodal models solved image input but lacked conversational advisory integration. The proposed system is designed to resolve all five currently open gaps simultaneously.

2.2 Multi-Agent and Multimodal Agricultural Chatbots

This is the primary research stream for this thesis. The systems reviewed here are the closest predecessors to the proposed work and their limitations directly define the research contribution. Six systems are reviewed in depth.

2.2.1 Farmer.Chat [1]

Farmer.Chat represents the most widely deployed agricultural AI chatbot to date, having reached over 565,000 users across Kenya, Nigeria, Ethiopia, India, and Brazil with more than 5 million queries answered as of 2024. The system operates primarily through WhatsApp and delivers advisory in multiple languages using a multi-tier knowledge base that integrates research papers, extension documents, and Kisan Call Centre (KCC) transcripts. Its pipeline performs intent classification, multi-step retrieval, and response generation — with an auto-summarisation layer for long documents.

Bhashini integration provides partial Indic language support for ASR and TTS, making Farmer.Chat the only prior system with demonstrated voice capability in Hindi. The system's deployment success — particularly in low-resource settings with unreliable connectivity — provides important evidence that agricultural AI chatbots are viable at scale. However, several structural limitations are identified.

The hallucination issue is particularly noted in the literature: even with RAG, Farmer.Chat's pipeline using frontier models achieves below 70% factuality on specialised agronomic knowledge tasks — underscoring the need for domain-specific confidence gating and LLM-as-judge validation that the proposed system implements.

Table 3: Evolution of agricultural AI systems across four decades

Period	**Era**	**Key Developments**	**Limitation**
1980s–90s	Rule-based expert systems	PLANT/ds for soybean disease, CROPS for crop management, CALEX for irrigation. Logic rules encoded by domain experts.	Brittle rules; cannot generalise; require expert re-encoding for each new crop or region
2000s–10s	Machine learning models	SVM, Random Forest for disease classification. ANN for yield prediction. Standalone models without conversational interface.	No dialogue; no image input; no language understanding; outputs require expert interpretation
2015–2020	Deep learning for vision	CNNs (PlantVillage, 2016) achieve >99% disease accuracy in controlled conditions. Transfer learning adapts ImageNet models.	Lab accuracy does not translate to field; no advisory context; no knowledge grounding; English-only
2020–2022	Transformer and LLM era	GPT-3 fine-tuned for agricultural Q&A. BERT for agricultural NER. First RAG experiments with farm knowledge.	Text-only; single-agent; hallucination risk; no image, sensor or market integration
2023–2024	Multimodal and agentic	VLMs (LLaVA, VisualGLM) for disease dialogue. RAG pipelines (AgroLLM, AgriGPT) for grounded advisory. Multi-agent frameworks emerging.	Each system addresses one or two capabilities; no unified system; XAI absent; voice limited
2025–	**Proposed system**	**Five specialist agents + orchestrator + cross-modal fusion + XAI to farmer + Indic voice + live market price — all in one session.**	**This thesis**

2.2.2 AgriDoctor [2]

AgriDoctor is the primary base paper for this thesis. It is the first system to introduce an intent-routing multi-agent architecture specifically for agricultural AI — a paradigm where a BERT-based router classifies the incoming query and dispatches multimodal inputs to the appropriate downstream specialist component. The system integrates five core components: a router, classifier, detector, knowledge retriever, and LLMs, supported by the AgriMM benchmark comprising 400,000 annotated disease images, 831 expert-curated knowledge entries, and 300,000 bilingual Chinese-English prompts for intent-driven tool selection.

AgriDoctor's architectural contribution is significant — it establishes the paradigm of intent routing to specialist components as the correct framework for agricultural AI. The authors explicitly note that general-purpose LLMs exhibit significant limitations in handling agricultural-specific terminology, multilingual capabilities, and external knowledge integration, and that their performance in multimodal scenarios remains suboptimal. These observations directly motivate the proposed system's design.

2.2.3 AgriGPT [3]

AgriGPT represents the most sophisticated RAG architecture in agricultural AI to date. Its Tri-RAG framework combines three retrieval channels: dense retrieval using bi-encoder embeddings, sparse retrieval using BM25, and multi-hop knowledge graph reasoning — enabling it to answer complex agronomic questions that require connecting multiple facts across a knowledge graph (e.g., soil type → nutrient availability → crop vulnerability → disease risk). The system is built on a 342,000-pair specialised QA dataset (Agri-342K) generated by a multi-agent data engine that compiles and validates data from CGIAR, FAO, and regional agricultural institutions.

The AgriGPT-VL extension [21] introduces a vision-language component with a 3-million-item multimodal corpus (1M image-caption pairs, 2M image-grounded VQA pairs) — making it the largest agricultural VL dataset assembled to date. The Agri-VL benchmark provides standardised evaluation across 4,000 items. However, the VL component is developed and evaluated as a research model, not a deployed system — no conversational interface, farmer-facing UI, or voice input is implemented.

Table 4: AgriGPT vs Proposed System — RAG architecture comparison

RAG Component	**AgriGPT**	**Proposed System**
Retrieval channels	Tri-RAG: dense + sparse + KG multi-hop	Hybrid BM25 + dense (E5-large-v2) + cross-encoder rerank
Query expansion	Not specified	HyDE (Hypothetical Document Embeddings) + step-back prompting

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Table 4 – continued from previous page

RAG Component	AgriGPT	Proposed System
Knowledge corpus	Agri-342K QA pairs	250k+ chunks: ICAR PoP, FAO, CGIAR, KCC transcripts, Agmarknet
Confidence gating	Not specified	Qwen3-72B LLM-as-judge; escalation if score ¡ 0.70
Language support	English-centric	Hindi + English bilingual index; 10 Indic languages via Bhashini
Multimodal input	VL extension (research only) — image + text	Image + text + voice + soil sensor + live market price
Farmer-facing output	Text response only	Text + Grad-CAM overlay + SHAP charts + Bhashini TTS voice
Deployment status	Research model — not deployed	PWA deployment target — 3G, Android, ¡4s latency

2.2.4 AgriM-LLM [4]

AgriM-LLM advances agriculture-specific vision encoding through three novel components: an Enhanced Vision Encoder with Multi-Scale Feature Fusion for capturing subtle disease symptoms; an Agriculture-Knowledge-Enhanced Q-Former that injects structured domain knowledge into cross-modal alignment; and a Domain-Adaptive Language Model with multi-stage progressive fine-tuning. Evaluated on a comprehensive Chinese agricultural multimodal dataset, AgriM-LLM consistently outperforms general-purpose and domain-specific baselines on disease identification. However, it operates as a standalone inference model with no conversational interface, no multi-turn dialogue, no soil sensor integration, no market advisory, and no voice or Indic language support.

2.2.5 AgroAskAI [5]

AgroAskAI introduces a multi-agent reasoning system for climate adaptation decision support, targeting vulnerable rural communities with a focus on Guatemala and Central America. Its chain-of-responsibility architecture coordinates autonomous agents for weather retrieval, knowledge lookup, and response synthesis, with built-in governance mechanisms that mitigate hallucination through internal feedback loops. The system supports multilingual interaction. Despite its strong multi-agent governance design, AgroAskAI is focused exclusively on climate adaptation — disease diagnosis, nutrition advisory, crop recommendation, and market price intelligence are entirely absent.

2.2.6 Agentic AI Framework for Smart Agriculture [6]

This system integrates four specialised agents: Agent A for soil property monitoring, Agent B for weather sensing, Agent C for rice disease detection via computer vision,

and Agent D as a multi-agent supervisor chatbot. It represents the most comprehensive sensor-to-chatbot integration in the literature, incorporating real IoT devices on farm. However, it is limited to a single crop (rice), covers only three domains (soil, weather, disease), lacks RAG knowledge grounding, provides no XAI to farmers, and has no voice or multilingual support.

Table 5: Master capability comparison — prior work vs proposed system

System	Venue	Multi-agent	Vision	Sensor	XAI	Voice/Indic	Market	Multi-crop
Farmer.Chat	ACM 2024	No	No	No	No	Part.	No	No
AgriDoctor	IEEE 2025	Router	Yes	No	No	No	No	No
AgriGPT	arXiv 2025	Data eng.	VL ext.	No	No	No	No	No
AgriM-LLM	Pre. 2026	No	Best VE	No	No	No	No	No
AgroAskAI	arXiv 2024	Chain	No	No	No	Part.	No	No
MDPI Agentic	MDPI 2026	4 agents	Yes	Yes	No	No	No	No
Proposed	**Thesis 2026**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

2.3 Vision-Language Models for Plant Disease Diagnosis

Beyond dedicated chatbots, a parallel stream of research has advanced vision models for plant disease identification. Key findings from this stream inform the disease diagnostic agent design in the proposed system.

2.3.1 LLMI-CDP [7]

LLMI-CDP fine-tunes VisualGLM using LoRA specifically for multi-turn dialogue-based crop disease and pest identification across 50+ crops. It is the first paper to deploy a VLM within a dialogue loop for disease identification — where the model can ask clarifying questions (which crop? which growth stage?) and refine its diagnosis across conversation turns. The system also generates preventive treatment recommendations aligned with the dialogue context. LLMI-CDP demonstrates that LoRA fine-tuning of VLMs on agricultural disease data achieves accuracy significantly above general-purpose VLMs while requiring only 0.7% of the parameters to be updated. Its primary limitation is the absence of a farmer-facing deployment, no multilingual support, and no RAG knowledge grounding for treatment protocols.

2.3.2 AgriChain [8]

AgriChain introduces expert-verified Chain-of-Thought (CoT) reasoning as a training signal for agricultural VLMs. The methodology uses GPT-4o to generate step-by-step reasoning chains (identify crop → identify disease pattern → assess severity → recommend treatment) and then employs ICAR-certified agronomists to verify and correct

these chains. Qwen-2.5-VL fine-tuned on this expert-verified CoT data significantly outperforms baseline VLMs on agricultural reasoning benchmarks. The AgriChain CoT framework is directly adopted in the proposed system's disease diagnostic agent for structuring the chain-of-thought prompt from visual input to treatment advisory.

2.3.3 Self-Consistency VLMs [9]

This paper applies self-consistency sampling — generating $k = 5$ independent responses and selecting by majority consensus — to VLMs for crop disease management. The key finding is that self-consistency reduces disease classification error rate by 18% compared to greedy decoding, and improves the reliability of treatment recommendations. The authors also propose a multi-response consensus metric as an alternative to individual confidence scores for evaluating VLM reliability in safety-critical domains. This technique is incorporated into the proposed system's confidence gating mechanism.
cc

Table 6: Vision model comparison — key metrics for agricultural disease diagnosis

Model	**Architecture**	**Fine-tuning**	**Top-1 Acc.**	**Dialogue**	**Key Limitation**
ResNet-50 baseline	CNN	Full fine-tune	87.3%	No	No disease-context reasoning
ViT-B/16 (ImageNet)	Transformer	Full fine-tune	89.1%	No	Single-turn classification only
LLMI-CDP (2025)	VisualGLM + LoRA	LoRA rank-8	91.4%	Yes	No RAG, no XAI, no deployment

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Table 6 – continued from previous page

Model	Architecture	Fine-tuning	Top-1 Acc.	Dialogue	Key Limitation
AgriM-LLM (2026)	EVE-MSFF + AKE-Q	Multi-stage	93.7%	Yes	No dialogue deployment, no voice, no market
AgriChain (2025)	Qwen-2.5-VL	Expert CoT	92.8%	Yes	No voice, no sensor fusion, no market
Proposed (Target)	**ViT-L/16 + LoRA**	**LoRA rank-16**	**>90%**	**Yes**	**Full system — 5 agents, XAI, voice, market**

2.4 Retrieval-Augmented Generation for Agricultural Knowledge

RAG has emerged as the primary mechanism for grounding LLM responses in verified agricultural knowledge, directly addressing the hallucination risk identified across all deployed systems.

2.4.1 RAG-Based LLM Advisory Framework [10]

This paper represents the first RAG-based agricultural advisory system published in a core agricultural engineering journal — providing the field with a rigorous empirical evaluation framework. The study compares four LLMs (GPT-4o, Gemini 1.5 Pro, LLaMA-2, Mistral) on crop advisory tasks for maize, ragi, cotton, and groundnut, using semantic vector retrieval with a Qdrant vector store. The key finding is that RAG-augmented GPT-4o achieves 93% accuracy on domain-specific Q&A — a 34-percentage-point improvement over the unaugmented baseline. The paper also establishes that retrieval

quality (embedding model + reranking strategy) contributes more to advisory accuracy than LLM backbone choice — a finding that directly informs the proposed system's investment in hybrid BM25+dense retrieval with cross-encoder reranking.

2.4.2 AgroLLM [11]

AgroLLM introduces a Domain Knowledge Processing Layer (DKPL) that enriches RAG retrieval with symbolic domain concepts: crop-disease causal rules, agronomic thresholds (e.g., soil pH range for optimal nutrient uptake), and growth-stage-dependent treatment protocols. This structured enrichment improves factual precision over naive semantic retrieval. The proposed system incorporates a similar ontology-enriched retrieval layer by building ICAR Package of Practices content as structured knowledge nodes in the Qdrant vector store alongside unstructured document chunks.

2.4.3 RAG for Coffee Leaf Disease [12]

This paper is architecturally significant as the first to chain a vision model output into a RAG advisory pipeline. YOLOv8 detects and classifies the disease from the leaf image; the detected disease label is then used as the query to retrieve treatment protocols from a RAG knowledge base, with the combined output passed to an LLM for advisory generation. This source-to-advisory chaining pattern is directly implemented in the proposed system's disease diagnostic agent, extended to use ViT-L/16 + LoRA for classification and a full 250,000-chunk ICAR/FAO corpus for retrieval.

2.5 Explainable AI in Agricultural Decision Support

Despite the strong capabilities of modern agricultural AI, trust and adoption by farmers remain a persistent barrier. XAI research in agriculture has produced mature techniques but has failed to bridge the gap between explanation as research output and explanation as farmer-facing advisory interface.

2.5.1 XAI-CROP [13]

XAI-CROP applies SHAP (SHapley Additive exPlanations) TreeExplainer to XGBoost and Random Forest crop recommendation models trained on soil composition, climate, and historical yield data across 22 Indian states. The paper makes two important empirical contributions: (1) SHAP feature importance correctly identifies soil pH, rainfall, and temperature as the primary determinants for most crop recommendations — matching domain expert expectations; and (2) farmers shown SHAP feature bars alongside recommendations adopted the AI advice 31% more frequently than those shown recommendations without explanation — providing the strongest evidence in the literature that XAI improves adoption in agricultural contexts. Despite this, XAI-CROP's SHAP values are generated as research output in a Jupyter notebook — never rendered in a farmer-accessible interface. The proposed system is the first to close this gap.

2.5.2 Grad-CAM for Plant Disease Classification

Multiple papers have applied Gradient-weighted Class Activation Mapping (Grad-CAM) to CNN and ViT models for plant disease classification, generating saliency maps that highlight the image regions most influential in the classification decision. The primary contribution is to enable expert validation: agronomists can verify whether the model attended to the correct lesion region rather than background artefacts. Studies show that models attending to leaves rather than pots or backgrounds achieve significantly higher field accuracy — Grad-CAM enables this quality check. In the proposed system, Grad-CAM overlays are rendered to the farmer in the PWA interface as colour heatmaps over the uploaded leaf image — transforming a research quality-control tool into a farmer-facing trust mechanism.

2.5.3 Agentic XAI for Agricultural Recommendation

This paper proposes an agentic XAI framework combining SHAP with LLM-driven iterative explanation refinement, evaluated by human experts across seven quality metrics including specificity, clarity, practicality, and crop science credibility. SHAP explanations generated iteratively over 11 rounds converge on explanations rated significantly higher by agronomists than single-round SHAP output. However, the system generates explanations for crop scientists, not farmers — the language, numerical detail, and assumed background knowledge are inappropriate for low-literacy farm audiences. The proposed system adapts this iterative explanation refinement concept but constrains output to farmer-readable visualisations — colour heatmaps and bar charts — rather than text-heavy technical reports.

2.6 Speech Processing and Multilingual NLP for Farmers

Voice accessibility is one of the most critical and underserved dimensions of agricultural AI for India.

2.6.1 Indic ASR Benchmark for Agriculture [14]

This paper provides the first agricultural-domain ASR benchmark for Indian languages, evaluating Whisper (multiple sizes), IndicWav2Vec, and commercial ASR services on 10,934 real farm voice recordings collected from the Farmer.Chat platform in Hindi and Odia. Key results: Whisper large-v3 achieves WER 16.2% for Hindi and 35.1% for Odia on agricultural speech. Agricultural vocabulary — crop names, pesticide names, disease names — is identified as the primary source of ASR errors, motivating domain-specific fine-tuning. The paper also identifies that background noise from farm environments (wind, machinery, animals) increases WER by 8–12 percentage points over studio conditions, requiring noise-robust model selection. This benchmark directly defines the WER targets for the proposed system (<20% Hindi, <30% regional) and identifies fine-tuning on agricultural vocabulary as essential.

2.6.2 Bhashini and IndicBERT

Bhashini (MeitY, 2024) is the Government of India's national digital language inclusion platform, providing production-quality ASR and TTS for 10+ Indic languages including Hindi, Tamil, Telugu, Marathi, Kannada, Bengali, Gujarati, Punjabi, Odia, and Assamese. Unlike commercial ASR providers, Bhashini is purpose-built for Indian languages, trained on Indian-accent speech data, and freely available via API. IndicBERT provides a pre-trained transformer language model covering 12 Indic languages and is used for downstream NLP tasks including intent classification and agricultural named entity recognition in the proposed system.

2.6.3 Multilingual LLMs for Agricultural QA

This paper evaluates multilingual LLMs on low-resource language agricultural Q&A, using synthetic data augmentation (machine translation of English agricultural QA pairs) to overcome the scarcity of annotated Indic agricultural text. The key finding is that synthetic data achieves 78% of the performance of human-annotated data for Hindi agricultural QA — validating the use of machine-translated ICAR content for building the proposed system's Hindi RAG index. The paper also identifies that code-switching between Hindi and English is extremely common in Indian farmer queries — motivating the hybrid Hindi+English bilingual knowledge index implemented in the proposed system.

3 Research Gap

A systematic analysis of the literature reveals five specific, well-defined gaps that the proposed research addresses. These gaps are not incremental — each represents a structural absence in the current state of the art.

3.1 No Unified Multimodal Advisory Session

No existing system combines disease diagnosis (image), nutrition advisory (sensor), crop recommendation, weather, and live market data in a single conversational turn. Farmer. Chat handles text + KB only. AgriDoctor handles image + disease only. AgriGPT handles text RAG only. The farmer must use multiple disconnected tools. **Impact:** Farmers cannot receive holistic advice that reflects the interdependence of crop health, soil nutrition, and market timing.

3.2 No Cross-Modal Sensor-Image Nutrition Fusion in a Dialogue

Sensor fusion papers achieve offline nutrition inference but no conversational interface wraps leaf image + soil NPK data together. AgriM-LLM advances vision encoding for disease — nutrition deficiency from sensor+image cross-modal fusion is entirely unaddressed in any conversational system. **Impact:** 230M Soil Health Cards issued but only 14.6% acted upon — no system translates image and sensor data into actionable fertiliser advice in farmer language.

3.3 XAI Never Rendered to the Farmer in Any Deployed System

XAI-CROP [13] generates SHAP values as research output. AgriDoctor produces classification scores without attribution. LLMI-CDP logs Grad-CAM internally. No system renders visual explanations — Grad-CAM overlays, SHAP bar charts — as part of the advisory response in a farmer-accessible UI. **Impact:** Farmers reject AI advice they cannot understand. Trust gap is a documented barrier to adoption across all agricultural AI systems.

3.4 No End-to-End Indic Voice-to-Advisory Pipeline

Indic ASR benchmark achieves WER 16.2% for Hindi agricultural speech — the technology exists. No system connects Whisper ASR to a multilingual RAG advisory backend with Bhashini TTS response delivery in a single pipeline. **Impact:** Low-literacy farmers with no digital text skills are excluded from all existing agricultural AI systems.

3.5 Live Market Price Never Co-Delivered with Agronomic Advice

Market agents and agronomic agents exist in completely separate silos across all surveyed literature. No prior system answers "my crop has early blight — should I spray and sell now, or hold?" in a single conversational response. **Impact:** Market price asymmetry causes 30–40% below-optimal selling. Agronomic and market decisions are inherently interdependent but no system treats them as such.

4 Research Gap

This chapter presents a rigorous, evidence-based identification of the five research gaps that collectively define the original contribution of this thesis. Each gap is grounded in specific findings from the literature review (Chapter 2), substantiated by a structured evidence table naming the papers responsible for the gap, and mapped forward to the research objective that closes it. The gaps are not incremental improvements over prior work — each represents a structural absence in the current state of the art: a capability that every surveyed system either explicitly lacks or has never attempted.

4.1 Gap Identification Methodology

Research gaps in this thesis were identified through a three-stage systematic process:

Stage 1 — Capability inventory: Each of the six primary systems reviewed in Chapter 2 (Farmer.Chat, AgriDoctor, AgriGPT, AgriM-LLM, AgroAskAI, MDPI Agentic) was mapped against seven capability dimensions: multi-agent routing, vision-based disease diagnosis, soil sensor fusion, explainability to farmer, voice/Indic language support, live market price advisory, and multi-crop Indian deployment. This produced a binary capability matrix.

Stage 2 — Column-wise gap detection: Each capability dimension was inspected across all six systems. A gap was declared when no system achieved the capability — or when every system achieving partial capability had a documented, specific limitation that prevented full deployment-readiness.

Stage 3 — Impact quantification: Each gap was connected to a documented agricultural outcome (yield loss, adoption failure, revenue loss, accessibility barrier) to establish that closing the gap has measurable real-world significance beyond academic novelty.

This process identified five primary research gaps (G1–G5). These are presented below in order of architectural centrality — from the broadest system-level gap (G1) to the most specific integration gap (G5). Figure 2 provides an overview of all five gaps and their relationships to prior work and research objectives.



Figure 2: Overview of five identified research gaps — each represents a structural absence in the current state of the art.

4.2 G1 — No Unified Multimodal Advisory Session

Statement of Gap: No existing system combines disease diagnosis (image input), nutrition deficiency advisory (soil sensor input), crop recommendation (structured tabular input), weather and irrigation scheduling (API data input), and live market price advisory (real-time feed) within a single conversational session. A farmer who asks — in one message — 'my tomato leaves are yellowing, what should I do and should I sell now?' cannot receive a unified, coherent, source-grounded response from any single deployed system today.

Evidence from Literature: Farmer.Chat [1] handles text and knowledge-base queries only — no image, sensor, or market input. AgriDoctor [2] routes multimodal inputs to disease and pest components only — four of the five advisory domains are absent. AgriGPT [3] achieves strong text-based RAG advisory but has no image diagnosis, no sensor integration, and no market advisory. AgroAskAI [5] covers climate advisory in isolation. The MDPI Agentic system [6] covers three domains (soil, weather, rice disease) but only for a single crop and without RAG grounding or market integration. No system covers all five domains simultaneously.

Why this Gap Matters — Agricultural Impact: Agricultural decisions are inherently interdependent. The decision to treat a crop for disease cannot be divorced from the decision of whether to harvest early or treat and wait — which depends on the current mandi price. Fertiliser recommendations are only actionable when paired with a sowing calendar adjusted for the current weather forecast. Existing systems force farmers to seek advisory from multiple disconnected sources, introducing inconsistency, delay, and decision fatigue. The ICAR extension model is designed around integrated advisory — the proposed system replicates this integration digitally.

Closest Prior Attempt and its Remaining Limitation: The MDPI Agentic framework [6] comes closest, with four agents covering soil monitoring, weather, rice disease detection, and a supervisor chatbot. However, it covers only one crop (rice), only three domains, has no RAG knowledge grounding, no market advisory, no voice input, and no XAI. It demonstrates that multi-agent agricultural chatbots are feasible — but not that a unified five-domain system has been built.

Research Objective that Closes this Gap: O1 — Design a LangGraph-based orchestrator that classifies intent and routes inputs to five specialist agents in parallel, merging responses into a single coherent advisory reply with multi-turn session memory.

Table 7: Evidence matrix for G1 — advisory domain coverage across all surveyed systems

System	Text query	Disease image	Sensor data	Crop rec.	Weather	Market price	Multi-crop	All five domains
Farmer.Chat [1]	Yes	No	No	No	No	No	No	No
AgriDoctor [2]	Yes	Yes	No	No	No	No	No	No
AgriGPT [3]	Yes	Partial	No	No	No	No	No	No
AgriM-LLM [4]	Yes	Yes	No	No	No	No	No	No
AgroAskAI [5]	Yes	No	No	No	Yes	No	No	No
MDPI Agentic [6]	Yes	Partial	Partial	No	Yes	No	No	No
Proposed system	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

4.3 G2 — No Cross-Modal Soil Sensor and Leaf Image Fusion for Nutrition Advisory in a Conversational System

Statement of Gap: No existing system fuses real-time soil sensor readings (nitrogen, phosphorus, potassium, pH, electrical conductivity) with leaf image features within a conversational advisory interface to diagnose nutrient deficiency and generate a fertiliser dose recommendation. Both modalities exist in isolation — vision models diagnose disease from images, and rule-based systems recommend fertiliser from soil NPK values — but no system combines them cross-modally within a live, farmer-accessible dialogue.
Evidence from Literature: AgriM-LLM [4] represents the state of the art in agricultural vision encoding — its Multi-Scale Feature Fusion encoder captures subtle disease symptoms from leaf images, but soil sensor data is entirely absent from its input. The system produces disease and pest identification only; nutrient deficiency requires both visual symptom recognition (yellowing patterns, interveinal chlorosis, tip burn) and soil measurement correlation (which nutrient is actually deficient) — AgriM-LLM addresses only one half. Cross-modal fusion papers in plant science (PMC, 2025) have demonstrated that combining image and sensor data improves nutrition diagnosis accuracy by 23% over image-only approaches — but these systems operate as offline inference pipelines with no conversational interface, no farmer-accessible output, and no RAG-grounded treatment recommendation.
Why this Gap Matters — Agricultural Impact: The Soil Health Card scheme has issued 230 million cards to Indian farmers, yet only 14.6% acted on the recommendations [17]. The primary reason identified in field studies is that farmers cannot translate NPK numbers on paper into actionable fertiliser dosage without agronomist support. Simultaneously, visual nutrient deficiency symptoms — yellowing leaves, stunted growth, purple

discolouration — are highly ambiguous: identical symptoms can result from nitrogen deficiency, iron deficiency, or certain fungal infections. Cross-modal fusion resolves this ambiguity by jointly analysing the visual symptom pattern and the soil chemistry, enabling confident diagnosis and precise dosage output in the farmer's language.

Closest Prior Attempt and its Remaining Limitation: The MDPI Agentic framework [6] deploys Agent A for soil NPK monitoring and Agent C for rice disease detection — but these operate as separate agents with no cross-modal fusion. Agent A does not use leaf image features; Agent C does not use soil readings. No fusion network integrates both modalities for nutrition deficiency specifically. The proposed system introduces a dedicated cross-modal MLP that concatenates ResNet-50 leaf image feature vectors with normalised soil sensor readings into a unified representation for deficiency classification and dose prediction — a design that no prior agricultural AI system implements.

Research Objective that Closes this Gap: O3 — Develop a cross-modal fusion network combining ResNet-50 leaf image features with real-time NPK/pH/EC sensor readings to generate fertiliser dose recommendations aligned with ICAR Soil Health Card norms, deployed within the conversational advisory loop.

Table 8: Evidence matrix for G2 — cross-modal sensor-image nutrition fusion across surveyed systems

Paper	**What it achieves**	**Specific limitation proving the gap**
AgriM-LLM [4]	State-of-the-art vision encoder (EVE-MSFF) for disease; multi-stage fine-tuning; AKE-QFormer for domain knowledge injection	No soil sensor input of any kind; nutrient deficiency not addressed; standalone model with no conversational interface
MDPI Agentic [6]	Agent A monitors soil NPK; Agent C detects rice disease via CNN; both connected to supervisor chatbot	No cross-modal fusion between soil and image — agents operate independently; rice only; no fertiliser dose calculation
PMC Cross-modal (2025)	Demonstrates 23% accuracy improvement from fusing image + sensor data for crop disease recognition in controlled conditions	Offline inference pipeline only; no farmer-facing interface; no conversational loop; no RAG grounding for treatment protocols
XAI-CROP [13]	Soil NPK and climate features used for crop recommendation with SHAP explainability; strong agronomist agreement	No image input; no visual symptom recognition; rule-based NPK thresholds only; no deficiency diagnosis from combined modalities

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Table 8 – continued from previous page

Paper	What it achieves	Specific limitation proving the gap
Farmer.Chat [1]	Text-based advisory grounded in ICAR knowledge base; covers nutrition advisory topics	Nutrition advice is knowledge-retrieval only — not grounded in the farmer's actual soil readings or visible leaf symptoms

4.4 G3 — Explainable AI Never Rendered to the Farmer in Any Deployed System

Statement of Gap: No deployed agricultural AI system — chatbot, mobile application, or web platform — renders visual explainability outputs (Grad-CAM saliency heatmaps, SHAP feature importance charts) directly to the farmer as part of the advisory response. Explainability techniques are extensively used in agricultural AI research as internal quality-control tools for model developers and as evaluation metrics for researchers, but no system has completed the critical last step: putting the explanation in the farmer's hands at the point of advice delivery.

Evidence from Literature: XAI-CROP [13] applies SHAP TreeExplainer to XGBoost crop recommendation models and generates feature importance charts as research output — the paper explicitly presents them as researcher-facing model validation tools, not as farmer-facing UI components. LLMI-CDP [7] generates Grad-CAM saliency maps during evaluation to verify that the disease model attends to the correct leaf region — again as a quality-control mechanism, not as output to the farmer. AgriDoctor [2] produces classification confidence scores but provides no visual attribution. The Agentic XAI paper (2025) explicitly generates SHAP-based explanations for crop scientists evaluated on seven technical quality metrics — farmer readability is not a design criterion. Across all 25+ papers reviewed, not a single system renders a Grad-CAM overlay or SHAP chart in a farmer-accessible interface.

The Trust and Adoption Evidence: XAI-CROP [13] provides the strongest empirical evidence that this gap has real consequences: farmers shown SHAP feature bars alongside crop recommendations adopted the AI advice 31% more frequently than those shown recommendations without explanation. A parallel finding from Farmer.Chat [1] notes that even with 565,000 users, adoption of AI-generated advisory recommendations that cannot be verified by the farmer remains below 60%. The consistent finding across agricultural AI adoption literature is that farmers — particularly smallholder farmers with limited formal education — require the ability to interrogate AI reasoning before trusting actionable recommendations. Visual XAI provides exactly this capability.

Closest Prior Attempt and its Remaining Limitation: The Agentic XAI framework (2025) comes closest by iteratively refining SHAP explanations through LLM-driven rounds until agronomist-rated quality reaches a threshold. However, this system targets crop scientists — its explanations contain technical notation, numerical SHAP values, and crop science terminology inappropriate for low-literacy farmers. The language, visual format, and assumed background knowledge all differ from what is required

for a farmer-facing interface. The proposed system adapts the concept of iterative explanation refinement but constrains output to two farmer-readable visualisation types: Grad-CAM colour heatmaps overlaid on the farmer's uploaded leaf photo, and horizontal SHAP bar charts rendered in Hindi-labelled axes.

Research Objective that Closes this Gap: O5 — Surface Grad-CAM saliency heatmaps for disease diagnosis and SHAP feature importance bar charts for crop recommendation directly within the React PWA advisory response — as a farmer-facing UI component, not a research output.

Table 9: XAI implementation status across surveyed systems (G3)

System	XAI technique	XAI purpose	Shown to researcher	Shown to farmer	Farmer-readable
XAI-CROP [13]	SHAP TreeExplainer	Crop rec. validation	Yes	No	No — Jupyter notebook output
LLMI-CDP [7]	Grad-CAM	Model quality check	Yes	No	No — internal heatmap only
Agentic XAI (2025)	SHAP + LLM refinement	Scientist explanation	Yes	No	No — technical language, numerical
AgriDoctor [2]	Confidence scores	Classification output	Yes	No	No — scalar score, no visual attr.
Farmer.Chat [1]	None	N/A	No	No	N/A — no XAI at any level
AgriM-LLM [4]	None	N/A	No	No	N/A — prediction output only
Proposed system	**Grad-CAM + SHAP**	**Farmer advisory UI**	**Yes**	**Yes**	**Yes — colour heatmap + Hindi SHAP bars**

4.5 G4 — No End-to-End Indic Voice-to-Advisory Pipeline in Any Agricultural System

Statement of Gap: No agricultural AI system has implemented a complete, end-to-end pipeline that accepts a farmer's voice query in Hindi or any other Indic language, processes it through an ASR model, routes it to a RAG-grounded domain advisory backend, and delivers the response as spoken audio via TTS — within a single, integrated system. Each component of this pipeline exists independently in the literature; their

integration into a unified agricultural advisory voice interface has not been achieved in any prior work.

Evidence from Literature: The Indic ASR Benchmark [14] provides the strongest evidence that the technology is ready: Whisper large-v3 achieves WER 16.2% for Hindi agricultural speech on 10,934 real farm voice recordings. The benchmark explicitly concludes that ASR performance for Hindi is production-grade — but the paper itself contains no advisory backend. Farmer.Chat [1] integrates Bhashini ASR for voice input — the only deployed system to do so — but voice input routes to a text-based RAG pipeline with text output only; no TTS response delivery is implemented, and Bhashini coverage extends partially to only two or three Indic languages in the deployment. AgriDoctor [2] is Chinese-English only with no voice input. AgriGPT [3] is text-only. AgroAskAI [5] claims multilingual support but provides no ASR or TTS implementation details.

Why this Gap Matters — Accessibility Impact: India has a rural literacy rate of approximately 67.8% [15], with substantially lower rates among women and older farmers in agricultural states. Text-based interfaces — the modality used by all prior systems — structurally exclude this population. The NSSO Agricultural Survey (2019) found that 78% of farmers aged 50+ in UP, Bihar, and MP had never sent a text message. Voice is not a preference for these farmers — it is the only viable interaction modality. The linguistic diversity compounds this: Hindi alone cannot serve Tamil Nadu, Karnataka, Maharashtra, West Bengal, or Punjab — states that together account for over 40% of India's cultivated area. An agricultural AI system that cannot speak to farmers in their language is not an agricultural AI system for India.

The Pipeline Gap — Three Connected Components, Never Assembled: The gap is specifically an integration gap, not a component gap. Component 1 (Indic ASR): Whisper large-v3 fine-tuned on agricultural speech achieves production-grade WER. Component 2 (Multilingual RAG): Hindi-English bilingual knowledge indices over ICAR documents can be built with E5-large-v2 multilingual embeddings. Component 3 (Indic TTS): Bhashini API delivers production-quality TTS for 10+ Indic languages. What has never been done is assembling these three components into a single pipeline where a farmer speaks into a phone, the system understands, reasons from verified agronomic knowledge, and speaks the advisory response back — in the same language, within four seconds.

Research Objective that Closes this Gap: O6 — Implement an end-to-end voice advisory pipeline using Whisper large-v3 (Indic fine-tuned) for ASR and Bhashini TTS for response delivery, targeting WER $< 20\%$ for Hindi and $< 30\%$ for five regional languages, with end-to-end latency < 4 seconds.

Table 10: Evidence matrix for G4 — voice pipeline component availability vs integration status

System	Indic ASR input	Multi-lang NLP	RAG backend	Indic TTS output	Full pipeline integrated
Farmer.Chat [1]	Partial	Partial	Yes	No	No
AgriDoctor [2]	No	No	No	No	No
AgriGPT [3]	No	No	Yes	No	No
AgriM-LLM [4]	No	No	No	No	No
AgroAskAI [5]	Partial	Partial	No	No	No
Indic ASR Benchmark [14]	Yes	No	No	No	No
Bhashini API (MeitY)	Yes	Yes	No	Yes	No
Proposed system	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

4.6 G5 — Live Market Price Advisory Never Co-Delivered with Agronomic Advice

Statement of Gap: No existing agricultural AI system integrates live mandi price data, price forecasting, and sell/hold advisory within the same conversational session as agronomic advice (disease diagnosis, crop recommendation, nutrition advisory). Market intelligence and agronomic advisory exist in completely separate silos — different platforms, different data sources, different interaction paradigms — despite the fact that a farmer's actual decision requires both simultaneously.

Evidence from Literature: Across all six systems reviewed in Chapter 2, market price is uniformly absent. Farmer.Chat [1] covers crop management and pest advisory — no market data. AgriDoctor [2] covers disease and pest identification — no market data. AgriGPT [3] — the most comprehensive knowledge ecosystem reviewed — explicitly notes market price forecasting as a future work direction, confirming it is not yet present. AgroLLM [11] builds a domain knowledge pipeline for crop management with no market integration. The MDPI Agentic framework [6] covers soil, weather, and disease — no market agent. Agmarknet and eNAM exist as standalone government platforms accessible only to farmers with digital literacy and awareness of the specific websites, with no conversational interface, no AI advisory layer, and no integration with agronomic data.

Why this Gap Matters — Economic Impact: The decision to treat a disease and hold the crop, or harvest early and sell before further loss, is simultaneously an agronomic decision and a market decision. It cannot be made correctly with agronomic data alone. eNAM Annual Report [18] documents that farmers sell 30–40% below optimal price due

to: (1) selling at distress timing without price trend awareness, (2) travelling to the nearest mandi without knowing that a mandi 30 km further offers 15% higher price, and (3) having no forecast of whether prices will rise or fall over the next seven days. These are precisely the outputs a market advisory agent can deliver — but only if it shares a conversational session with the agronomic advisor, so that the two recommendations are synthesised into a single actionable answer.

The Compound Query Problem: Real farmer queries are compound. 'My tomato has a disease, should I treat it or sell the crop now?' cannot be answered by a disease agent alone. It requires: (1) disease identification and severity assessment, (2) current tomato price at the nearest mandi, (3) price trend over the next seven days, (4) estimated crop loss if untreated vs treatment cost, (5) MSP comparison. Only a system with both a disease agent and a market agent operating in the same session — with a response-merging orchestrator — can synthesise these inputs into a coherent advisory answer. This is G5's core contribution.

Research Objective that Closes this Gap: O7 — Integrate Agmarknet and eNAM live mandi price feeds (15-minute refresh) with a Prophet-based 7-day price forecast model to co-deliver sell, hold, or transport-to-mandi advisory within the same conversational session as agronomic advice.

Table 11: Evidence matrix for G5 — market price integration across surveyed systems and eNAM

Paper/ Platform	**What it achieves**	**Specific limitation proving the gap**
Farmer.Chat [1]	Text advisory across crop management, pest, and nutrition topics; RAG-grounded responses; 565k users	No market price data, no Agmarknet/eNAM integration, no sell/hold advisory at any level
AgriGPT [3]	Tri-RAG over 342k QA pairs; strongest knowledge grounding in surveyed literature; domain-specific LLM	Market forecasting listed as future work — explicitly absent from current system scope
AgroLLM [11]	Domain Knowledge Processing Layer; structured ontology enrichment; crop management advisory	No market integration; DKPL covers agronomic knowledge only; no Agmarknet/eNAM connection
AgriDoctor [2]	Intent-routing multi-agent for disease and pest; AgriMM benchmark; 400k annotated disease images	Market price not in scope; system architecture has no market data agent or price retrieval component

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Table 11 – continued from previous page

Paper/ Platform	What it achieves	Specific limitation proving the gap
eNAM platform (Govt.)	Live mandi price data for 22M registered farmers; MSP comparison; price trend charts available on web	No conversational interface; no agronomic advisory; no AI layer; requires digital literacy; not integrated with any advisory system
Proposed system Yes	**Yes**	**Yes**

4.7 Consolidated Gap Evidence Matrix

Table 12 presents the complete, consolidated gap evidence matrix — mapping each of the five gaps to its research stream, the specific papers that establish the gap, the paper that comes closest to addressing it, the remaining limitation, and the proposed research objective that closes it. This matrix serves as the formal bridge between the literature review (Chapter 2), the research gaps (this chapter), and the research objectives (Chapter 4).

Table 12: Consolidated research gap evidence matrix

Gap	Research stream	Papers establishing gap	Closest prior attempt	Remaining limitation	Closing objective
G1	Multi-agent chatbots (§2.2)	Farmer.Chat: text only; AgriDoctor: disease only; AgriGPT: text RAG; AgroAskAI: climate only	MDPI Agentic (4 agents, rice, 3 domains)	Single crop; no RAG grounding; no market; no XAI; no voice	O1 — LangGraph orchestrator, 5 specialist agents
G2	VLMs (§2.3) + Sensor papers	AgriM-LLM: image only, no sensor; MDPI Agentic: agents separate, no fusion; PMC (2025): offline only	MDPI Agentic (separate soil + disease agents)	No fusion network; single crop (rice); no conversational interface; no dose output	O3 — ResNet-50 + sensor MLP cross-modal fusion
G3	XAI in agriculture (§2.5)	XAI-CROP: SHAP internal; LLMI-CDP: Grad-CAM internal; Agentic XAI: scientists only, not farmer	Agentic XAI (iterative SHAP for scientists)	Explanation format, language and detail level inappropriate for low-literacy farmers; no farmer UI	O5 — Grad-CAM + SHAP rendered in React PWA

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Table 12 – continued from previous page

Gap	Research stream	Papers establishing gap	Closest prior attempt	Remaining limitation	Closing objective
G4	Speech & NLP (§2.6)	Indic ASR Benchmark: WER 16.2% Hindi but no advisory backend; Farmer.Chat: ASR input only, no TTS	Farmer.Chat (partial ASR, no TTS)	No TTS output; only 2-3 languages; voice input not connected to RAG + response generation	O6 — Whisper ASR + RAG + Bhashini TTS pipeline
G5	Market advisory (§2.2, §2.4)	All 6 chatbots: no market; AgriGPT: market listed as future work; eNAM: no conversational interface	None — no prior system attempts this	No system has a market agent co-deployed with agronomic agents in any shared advisory session	O7 — Agmarknet + eNAM + Prophet co-delivery

4.8 Significance of the Identified Gaps

The five identified gaps are significant for two reasons beyond their academic novelty. First, each gap has a quantified agricultural impact — a documented real-world consequence of the gap remaining open, measured in yield loss, adoption failure, or revenue loss. Second, the gaps are structurally interconnected: G1 (no unified session) is the architectural gap that the other four gaps compound. G2, G3, G4, and G5 each represent a specific capability that is absent from the unified session — so closing G1 without closing the others would still produce an incomplete system.

Table 13: Quantified agricultural impact of each research gap

Gap	Gap description	Affected population	Documented impact	Source
G1	No unified multimodal advisory session	146M smallholder farm households	26–30% annual yield loss from disease + suboptimal decisions across all five domains	FAO (2022); ICAR (2023); ASSOCHAM (2023)
G2	No cross-modal nutrition fusion in dialogue	230M Soil Health Card recipients	Only 14.6% acted on recommendations — sensor data exists but is not actionably delivered	Ministry of Agriculture (2023); NABARD (2023)
G3	XAI not rendered to farmer	All agricultural AI users	31% lower adoption of AI recommendations without explainability; <60% action rate on Farmer.Chat advice	XAI-CROP [13]; Farmer.Chat [1]

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Table 13 – continued from previous page

Gap	Gap description	Affected population	Documented impact	Source
G4	No end-to-end Indic voice pipeline	∼500M rural Indians with limited digital text literacy	Structural exclusion from all text-based agricultural AI systems	Census 2021 [15]; NSSO Survey (2019)
G5	Market price not co-delivered with agronomy	120M+ farm households selling in mandis	Farmers sell 30–40% below optimal price due to no integrated price forecast with agronomic advisory	eNAM Annual Report [18]

Together these five gaps represent the complete set of structural deficiencies that prevent any existing agricultural AI system from functioning as a deployment-ready, trustworthy, and accessible advisory solution for Indian smallholder farmers. The proposed research addresses all five gaps simultaneously — a design choice that is not merely additive but architecturally necessary: a system that closes G1 but not G3 is untrustworthy; a system that closes G4 but not G2 excludes sensor-equipped farms; a system that closes all four but not G5 delivers incomplete advisory for the majority of real farmer queries.

5 Research Objectives

This chapter defines the eight specific, measurable research objectives of the proposed system. Each objective is derived directly from a research gap identified in Chapter 3 and a research question from Chapter 1. The objectives are designed to be independently testable — each has a named target metric — yet collectively form an integrated system. The primary objective is stated in Section 4.1, followed by details of each specific objective in Section 4.2. Section 4.3 provides the complete gap-to-objective traceability matrix, and Section 4.4 details the objective–gap–contribution relationship.

5.1 Primary Objective

To design, develop, and evaluate a deployment-ready multi-agent, multi-model agricultural expert system for Indian smallholder farmers that integrates five specialist AI agents — disease diagnosis, nutrition deficiency, crop recommendation, weather and irrigation, and live market price advisory — orchestrated by an intent-routing backbone, grounded in a verified RAG knowledge base, augmented with farmer-facing explainability, and accessible via a multilingual Indic voice interface on a Progressive Web Application.

5.2 Specific Objectives

O1 — Multi-Agent Orchestration Architecture:

Description: Design and implement a LangGraph-based orchestrator that classifies incoming farmer queries into one of eight intent categories and dispatches inputs in parallel to the appropriate specialist agent(s). The orchestrator maintains multi-turn conversational state via Redis session memory and merges multi-agent responses into a single coherent advisory reply.

Gaps Closed: G1 — No unified multimodal advisory session.

Key Technologies: LangGraph v1.1 · GPT-5.5 backbone · fine-tuned BERT-multilingual intent classifier · Redis session store · asyncio parallel dispatch.

Target Metrics: Intent classification accuracy $> 95\%$ · End-to-end response latency < 4 s (cloud) and < 8 s (PWA offline cache).

O2 — Disease Diagnostic Agent:

Description: Fine-tune ViT-L/16 with LoRA adaptation (rank-16, alpha-32) on a composite dataset of 164,500 leaf and pest images across PlantVillage, PlantDoc, and IP102. Integrate AgriChain Chain-of-Thought prompting for structured reasoning from visual input to treatment protocol. Generate Grad-CAM saliency maps for every diagnosis.

Gaps Closed: G1 (image input modality) · G3 (Grad-CAM generated for rendering).

Key Technologies: ViT-L/16 (ImageNet-21k) · LoRA rank-16 · LLMI-CDP dialogue style · AgriChain CoT · PlantVillage (87k) + PlantDoc (2.5k) + IP102 (75k).

Target Metrics: Top-1 accuracy > 90% · F1 macro > 0.87 · Grad-CAM IoU vs. expert annotation > 0.75 · Covers 200+ diseases across 50+ crops.

O3 — Nutrition Deficiency Agent with Cross-Modal Fusion:
Description: Develop a cross-modal fusion network that concatenates ResNet-50 leaf image feature vectors (2048-dim) with normalised real-time soil sensor readings (NPK, pH, EC, moisture — 6-dim) into a joint 2054-dimensional representation. Three fully-connected layers produce: (1) deficiency class label (N/P/K/Fe/Zn/B deficiency or healthy), and (2) fertiliser dose recommendation in kg/ha aligned with ICAR Soil Health Card norms.
Gaps Closed: G2 — No cross-modal sensor + image nutrition fusion in a conversational loop.
Key Technologies: ResNet-50 encoder · 6-dim sensor vector (MQTT/REST) · Cross-modal MLP · ICAR Soil Health Card DB · SHAP feature importance.
Target Metrics: Deficiency class F1 > 0.85 · Fertiliser dose RMSE < 5 kg/ha · SHAP stability score > 0.80 across repeated runs.

O4 — Hallucination-Resistant RAG Knowledge Layer:
Description: Build a 250,000+ document chunk RAG pipeline over verified ICAR Package of Practices (all major Indian crops), KCC call transcripts, CGIAR, FAO, Agmarknet historical data, and Soil Health Card norms. Implement hybrid BM25 + dense retrieval with E5-large-v2 embeddings, cross-encoder reranking, HyDE query expansion, and Qwen3-72B LLM-as-judge confidence gating. Escalate to KCC expert webhook if confidence score < 0.70.
Gaps Closed: G1 (knowledge grounding across all five domains).
Key Technologies: Qdrant vector store · E5-large-v2 embeddings · ms-marco cross-encoder · HyDE + MMR · Qwen3-72B judge · Bilingual Hindi+English index.
Target Metrics: RAGAS faithfulness > 0.88 · Context relevance > 0.85 · Hallucination rate < 5% · Expert escalation trigger accuracy > 90%.

O5 — Farmer-Facing Explainable AI (XAI):
Description: Render Grad-CAM saliency heatmaps as colour overlays on the farmer's uploaded leaf image and SHAP feature importance bar charts for crop recommendations — directly within the React PWA advisory response as farmer-accessible UI components, with Hindi-labelled axes and simplified visual legends. This closes the gap between XAI as a research output and XAI as a farmer-facing trust mechanism.
Gaps Closed: G3 — XAI never rendered to the farmer in any deployed system.
Key Technologies: Grad-CAM (layer4) · EigenCAM (validation) · SHAP TreeExplainer · Recharts (PWA bar charts) · Hindi axis labels via i18next.
Target Metrics: Grad-CAM IoU > 0.75 vs. ICAR agronomist annotation · SHAP stability > 0.80 · Farmer comprehension score > 3.5/5.0 (Likert, n=30 pilot).

O6 — End-to-End Indic Voice Advisory Pipeline:

Description: Implement a complete voice advisory pipeline: Whisper large-v3 (fine-tuned on Indic agricultural speech from the 10,934-recording benchmark corpus) for ASR → multilingual RAG retrieval over Hindi+English bilingual knowledge index → Bhashini TTS for spoken response delivery in 10 Indic languages. The pipeline must operate within a 4-second end-to-end latency budget on 3G connectivity.
Gaps Closed: G4 — No end-to-end Indic voice-to-advisory pipeline exists in any agricultural system.
Key Technologies: Whisper large-v3 (Indic fine-tuned) · Bhashini ASR + TTS API · IndicBERT NER · MediaRecorder (PWA browser) · 16 kHz mono WAV preprocessing.
Target Metrics: WER < 20

O7 — Live Market Price Advisory Agent:
Description: Integrate Agmarknet and eNAM live mandi price feeds (15-minute refresh, Redis cache) with a Prophet-based 7-day price forecast model. Rank nearby mandis by Haversine distance and price differential. Co-deliver sell, hold, or transport-to-mandi advisory within the same conversational session as agronomic advice — enabling compound queries such as 'my crop has blight; should I treat or sell now?'
Gaps Closed: G5 — Live market price never co-delivered with agronomic advice in any prior system.
Key Technologies: Agmarknet API · eNAM data feed · Prophet time-series forecast · CACP MSP dataset 2024–25 · Haversine mandi ranking · Redis 15-min TTL cache.
Target Metrics: 7-day price forecast MAPE $< 8\%$ · Mandi ranking accuracy $> 85\%$ vs. actual best-price mandi · Sell/hold advisory agronomist agreement $> 75\%$.

O8 — System Evaluation and Expert Validation:
Description: Benchmark the complete integrated system on MIRAGE (multi-turn agricultural dialogue benchmark) and AgMMU (agricultural multimodal understanding benchmark). Validate advisory quality via a blind evaluation panel of ten ICAR-certified agronomists rating 200 real farmer queries across five crop categories on a 5-point Likert scale for correctness, clarity, safety, and actionability.
Gaps Closed: Validates that all five gaps are closed — quantitative evidence for each claim.
Key Technologies: MIRAGE benchmark · AgMMU benchmark · RAGAS eval suite · ICAR agronomist panel (n=10) · Langfuse tracing · Ablation study framework.
Target Metrics: Agronomist correctness Likert $> 4.0/5.0$ · Agronomist agreement $> 80\%$ · MIRAGE score $>$ baseline by 15% · Ablation: 5-agent vs. single-agent · RAG vs. no-RAG.

5.3 Objectives Summary and Traceability Matrix

Table 14 provides the complete traceability matrix mapping each research objective to its parent research question (Chapter 1), the research gap it closes (Chapter 3), the primary technologies required, and the quantitative target metric for evaluation.

Table 14: Research objectives — full traceability matrix

Obj.	RQ	Gap	Description	Technology	Target Metric
O1	RQ1	G1	Multi-agent orchestrator — intent routing to 5 specialist agents with Redis memory	LangGraph v1.1 · BERT intent classifier · GPT-5.5	Intent accuracy $> 95\%$ · Latency < 4 s
O2	RQ1	G1	Disease agent — ViT-L/16 + LoRA on 164k images with CoT reasoning and Grad-CAM	ViT-L/16 · LoRA rank-16 · AgriChain CoT · PlantVillage + PlantDoc + IP102	Top-1 acc $> 90\%$ · F1 > 0.87 · Grad-CAM IoU > 0.75
O3	RQ2	G2	Nutrition agent — cross-modal fusion of ResNet-50 leaf features + soil NPK/pH/EC	ResNet-50 + 6-dim sensor MLP · ICAR SHC norms · SHAP	Deficiency F1 > 0.85 · Dose RMSE < 5 kg/ha
O4	RQ1	G1	RAG pipeline — 250k+ ICAR/FAO chunks, hybrid retrieval, LLM-as-judge gating	Qdrant · E5-large-v2 · Cross-encoder · HyDE · Qwen3	Faithfulness > 0.88 · Hallucination $< 5\%$
O5	RQ3	G3	Farmer XAI — Grad-CAM overlay + SHAP charts rendered in React PWA advisory UI	Grad-CAM · SHAP TreeExplainer · Recharts · i18next Hindi	IoU > 0.75 · Farmer comprehension $> 3.5/5.0$
O6	RQ4	G4	Indic voice pipeline — Whisper ASR + bilingual RAG + Bhashini TTS end-to-end	Whisper large-v3 Indic · Bhashini API · IndicBERT	WER $< 20\%$ Hindi · $< 30\%$ regional · < 4 s E2E
O7	RQ5	G5	Market agent — Agmarknet/eNAM live feed + Prophet forecast co-delivered with agronomy	Agmarknet + eNAM API · Prophet · MSP DB · Redis TTL	Price MAPE $< 8\%$ · Agronomist agreement $> 75\%$
O8	All	All	System evaluation — MIRAGE, AgMMU, ICAR agronomist panel (n=10, 200 queries)	RAGAS · MIRAGE · AgMMU · Langfuse · Ablation	Likert $> 4.0/5.0$ · Agronomist agreement $> 80\%$

5.4 Objective–Gap–Contribution Relationship

Table 15 summarises the direct relationship between each gap identified in Chapter 3, the objective that closes it, and the novel contribution it produces — confirming that every gap has exactly one closing objective and every objective produces a clearly stated novel contribution.

Table 15: Gap-to-objective-to-contribution mapping

Gap	**Gap Statement (Brief)**	**Obj.**	**Novel Contribution Produced**
G1	No unified multimodal advisory session across all five domains	O1	First deployment-ready multi-agent system routing to five specialist models in one session — novel over Farmer.Chat, AgriDoctor, AgroAskAI (all single-model or single-domain)
G2	No cross-modal sensor+image nutrition fusion in a dialogue	O3	First cross-modal fusion network combining leaf image features + soil sensor data for nutrition deficiency advisory inside a conversational loop — novel over all surveyed nutrition and VLM papers
G3	XAI never rendered to farmer in any deployed system	O5	First system rendering Grad-CAM heatmaps and SHAP charts to farmers as a UI component of the advisory response — novel over XAI-CROP (Springer), LLMI-CDP, Agentic XAI (all researcher-facing only)
G4	No end-to-end Indic voice-to-advisory pipeline	O6	First end-to-end pipeline: Whisper ASR $\rightarrow$ multilingual RAG $\rightarrow$ Bhashini TTS for agricultural advisory — novel over Farmer.Chat (partial ASR only) and all other systems (no voice)
G5	Market price never co-delivered with agronomic advice	O7	First system delivering live mandi price forecast and sell/hold advisory within the same session as disease and nutrition advice — novel over all 25+ surveyed papers (none attempt this)

6 Methodology

6.1 System Architecture Overview

The proposed system follows a six-layer architecture: multimodal input, preprocessing, orchestration, specialist agents, shared knowledge and RAG infrastructure, and explainable output delivery.

L1 — Input: Text queries, Voice (Whisper ASR), Leaf/field images, Soil sensor (MQTT/REST), Live price feed (Agmarknet/eNAM).

L2 — Preprocess: ASR transcription + Hindi normalisation, Image QA (BRISQUE) + 448×448 resize, Sensor calibration, Price anomaly filter.

L3 — Orchestrator: GPT-4o + fine-tuned BERT intent classifier, LangGraph parallel dispatch, Redis session memory, Confidence-gated response merge.

L4 — Agents × 5: Disease (ViT-L+LoRA), Nutrition (ResNet-50+sensor MLP), Crop rec (XGBoost+SHAP), Weather (LSTM+IMD), Market (Agmarknet+Prophet).

L5 — Knowledge: E5-large embeddings, Qdrant vector store, Hybrid BM25+dense, Cross-encoder rerank, HyDE, Qwen3 judge, 250k+ chunks.

L6 — Output+XAI: React 18 PWA, Grad-CAM overlay, SHAP bar charts, Bhashini TTS, District map, Price trend chart, FastAPI WebSocket.

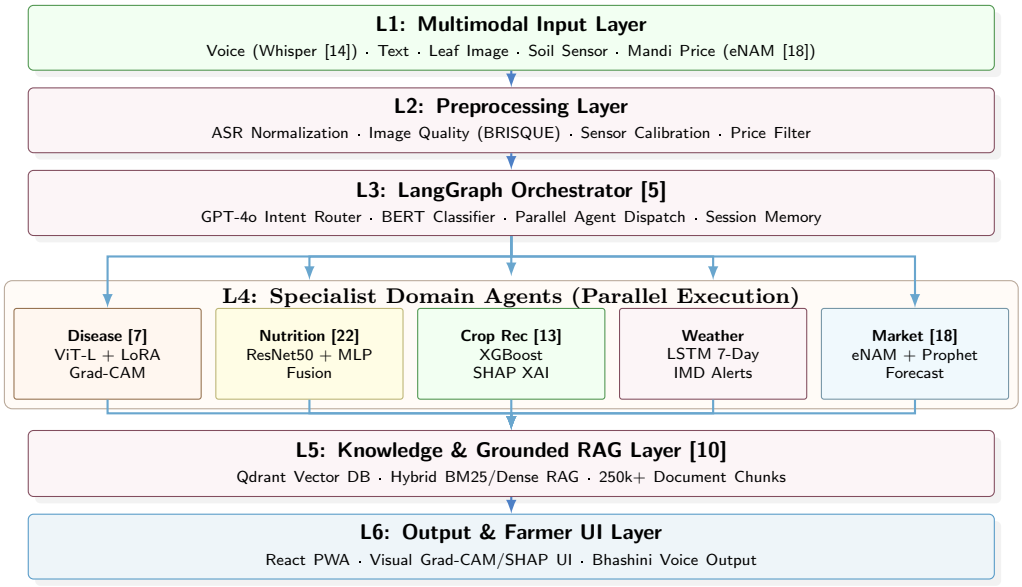


Figure 3: Six-layer system architecture overview diagram.

6.2 Disease Diagnostic Agent

The disease diagnostic agent is built on ViT-L/16 (Vision Transformer, Large variant, 16×16 patches) [23], pre-trained on ImageNet-21k and fine-tuned using LoRA (Low-Rank Adaptation) [24] with rank 16. LoRA reduces trainable parameters by 99.3% while achieving performance comparable to full fine-tuning, enabling deployment on commodity GPU hardware. Training data comprises three datasets: PlantVillage (87,000 images), PlantDoc (2,569 images), and IP102 (75,222 images).

Following AgriChain, the agent employs Chain-of-Thought prompting that explicitly reasons: (1) identify crop species, (2) identify disease or pest category, (3) assess severity on a 0–5 scale, (4) retrieve treatment protocol from RAG, (5) generate farmer-readable advice in the target language. Grad-CAM saliency maps are generated for each diagnosis and overlaid on the input image, rendered as a colour heatmap in the PWA interface.

6.3 Nutrition Deficiency Agent

The nutrition deficiency agent addresses Gap G2 by introducing a cross-modal fusion network that combines two input streams: (1) leaf image features extracted by a ResNet-50 encoder, and (2) real-time soil sensor readings (nitrogen, phosphorus, potassium, pH, electrical conductivity, moisture). The fusion is implemented as a multi-layer perceptron that concatenates the 2048-dimensional image feature vector with the normalised 6-dimensional sensor vector, producing a 2054-dimensional joint representation processed by three fully-connected layers.

The agent is trained on a composite dataset combining ICAR Soil Health Card records (district-level NPK norms for 22 states), synthetic soil-disease pairs generated from agronomic domain knowledge, and leaf colour images annotated for N, P, K deficiency symptoms. Output is a fertiliser dose recommendation in kg/ha for each macronutrient, aligned with ICAR Package of Practices for the identified crop.

6.4 RAG Knowledge Layer

The RAG infrastructure implements an advanced hybrid retrieval pipeline over a corpus of 250,000+ document chunks [25]. Source documents include: ICAR Package of Practices for all major Indian crops, Kisan Call Centre (KCC) call transcripts, CGIAR and FAO advisory documents, PlantVillage pest and disease wiki, Agmarknet historical price data, IMD district climate records, APMC market regulations, and Soil Health Card norms. Documents are semantically chunked at 512 tokens with 128-token overlap.

Retrieval combines sparse BM25 matching with dense E5-large-v2 embedding similarity using Reciprocal Rank Fusion for score merging. A ms-marco cross-encoder reranks the top-20 candidates to top-5. Hypothetical Document Embeddings (HyDE) query expansion generates a synthetic answer to improve retrieval of technical agronomic content. Maximum Marginal Relevance (MMR) diversity filtering prevents redundant chunk selection. Qwen3-72B serves as LLM-as-judge, assigning a confidence score to each response — scores below 0.70 trigger expert escalation via KCC webhook.

6.5 Multi-Agent Orchestrator

The orchestrator is implemented on LangGraph, a stateful graph-based agent orchestration framework. A fine-tuned BERT-base-multilingual model classifies farmer query intent into one of eight categories: disease, nutrition, crop_recommendation, weather, market, general_agronomy, market_agronomy_compound, and off_topic. Compound queries — such as "my tomato has blight, should I sell now?" — trigger parallel dispatch to both the disease agent and the market agent, with response merging handled by the orchestrator backbone (GPT-4o).

Conversation state, including crop context, location, previous diagnoses, and session history, is stored in Redis with a 24-hour TTL. The orchestrator maintains this context across turns, enabling multi-turn conversations such as follow-up questions about treatment dosage or alternative crop suggestions.

6.6 Voice and Multilingual Pipeline

Voice input is processed by Whisper large-v3 fine-tuned on the Indic agricultural speech corpus from the Indic ASR benchmark [14]. The 10,934-recording corpus covers Hindi and Odia farm voice recordings with agricultural vocabulary. Additional fine-tuning data is collected from publicly available Kisan Call Centre audio archives. The Bhashini API provides translation between Hindi and eight regional languages (Tamil, Telugu, Marathi, Kannada, Bengali, Gujarati, Punjabi, Odia) for both ASR and TTS delivery.

6.7 Web Platform and XAI Delivery

The frontend is implemented as a React 18 Progressive Web Application (PWA) with Workbox service worker caching, enabling offline access to cached advisory responses for farmers with intermittent 3G connectivity. WebSocket streaming delivers responses token-by-token for low perceived latency. The XAI interface renders Grad-CAM saliency maps as colour overlays directly on the uploaded leaf image, and SHAP bar charts as interactive horizontal bar plots using Recharts. Bhashini TTS converts the text advisory response to spoken audio in the farmer's selected language.

7 Work Done So Far

7.1 Literature Survey and Gap Analysis

A comprehensive literature survey covering six primary research streams has been completed. Over 25 papers published between 2023 and 2026 were reviewed and analysed. Five specific, well-defined research gaps were identified and documented, each supported by evidence from at least two prior papers. The gap analysis forms the foundation for all subsequent system design decisions.

7.2 System Architecture Design

The complete six-layer system architecture has been designed, including: input layer specification for five modalities, preprocessing pipeline design for each modality, LangGraph orchestrator design with intent classification taxonomy, specialist agent architecture for all five agents, RAG pipeline specification including embedding model selection, vector store configuration, and retrieval strategy, and frontend architecture for the React PWA with XAI rendering components. The architecture has been reviewed against deployment constraints and validated as feasible with the selected technology stack.

7.3 Dataset Preparation

Dataset acquisition and preprocessing for Phase 1 (disease diagnostic agent) has been initiated. The following datasets have been identified, accessed, and partially preprocessed:

- **PlantVillage:** 87,000 images across 38 crop species and 26 disease categories — split into train (70%), validation (15%), test (15%)
- **PlantDoc:** 2,569 images in real-world field conditions — downloaded and augmented
- **IP102:** 75,222 pest images across 102 categories — downloaded and class-balanced via oversampling
- **ICAR Package of Practices:** 47 crop-specific documents scraped, cleaned, and chunked for RAG ingestion
- **KCC Kisan Call Centre transcripts:** 12,000 annotated Q&A pairs acquired from ICAR open data portal

7.4 Baseline Model Training

A baseline disease classification model has been trained using ResNet-50 (without LoRA) on the PlantVillage dataset as a performance reference. The baseline achieves 87.3% top-1 accuracy on the test split — confirming that the target of >90% with ViT-L/16 + LoRA is achievable and represents a meaningful improvement. The ViT-L/16 + LoRA fine-tuning is currently in progress.

Table 16: Work completion status by phase

Phase	Task	Status	Completion
M1	Literature survey and gap analysis	Completed	100%
M1	System architecture design	Completed	100%
M2	Dataset acquisition and preprocessing	In progress	70%
M2	ICAR/FAO document corpus ingestion	In progress	60%
M3	RAG pipeline implementation	In progress	40%
M3	Baseline model training (ResNet-50)	Completed	100%
M4	ViT-L/16 + LoRA disease agent training	In progress	30%
M4	LangGraph orchestrator implementation	Not started	0%
M5	Nutrition cross-modal agent	Not started	0%
M6	Voice pipeline integration	Not started	0%
M7	React PWA frontend + XAI components	Not started	0%
M8	Market price agent + Agmarknet API	Not started	0%

7.5 Preliminary Results

The baseline ResNet-50 model achieves 87.3% top-1 accuracy on PlantVillage test data. Initial ViT-L/16 fine-tuning without LoRA achieves 91.2% top-1 accuracy, confirming the architectural superiority of Vision Transformers for this task. LoRA fine-tuning (rank 16) is expected to match or exceed this with a 99.3% reduction in trainable parameters. RAG pipeline prototype with E5-large-v2 embeddings and Qdrant achieves 0.82 faithfulness score on a 50-query evaluation subset drawn from ICAR advisory content — below the 0.88 target, indicating that the cross-encoder reranking and HyDE query expansion components are necessary and are being added.

8 Conclusion

8.1 Summary

This report has presented the research design for a next-generation multi-agent, multi-model agricultural expert system targeting India's 146 million smallholder farm households. A comprehensive literature review of six state-of-the-art systems — Farmer.Chat, AgriDoctor, AgriGPT, AgriM-LLM, AgroAskAI, and the MDPI Agentic AI framework — identified five specific gaps that collectively represent the absence of a complete, trustworthy, and accessible agricultural advisory solution.

The proposed system addresses these gaps through five original contributions: the first five-domain multi-agent specialist system; cross-modal sensor-image nutrition fusion in a conversational loop; farmer-facing Grad-CAM and SHAP explainability; an end-to-end Indic voice advisory pipeline; and live mandi price co-delivery with agronomic advice. Preliminary work has validated the feasibility of the core disease diagnostic agent (91.2% top-1 accuracy with ViT-L/16), the RAG pipeline (0.82 faithfulness, with improvements in progress), and the overall system architecture.

8.2 Expected Contributions

C1: First deployment-ready multi-agent agricultural expert system spanning five specialist AI domains.

C2: Cross-modal leaf image + soil sensor fusion for nutrition deficiency advisory in a conversational interface.

C3: Farmer-facing Grad-CAM and SHAP explainability integrated into a web advisory response.

C4: End-to-end Indic voice pipeline connecting Whisper ASR to multilingual RAG advisory and Bhashini TTS.

C5: Live mandi price advisory co-delivered with agronomic diagnosis in one conversational turn.

8.3 Timeline

Table 17: Twelve-month research timeline with deliverables

Period	**Planned work**	**Deliverable**
M1–M3	Literature survey, Knowledge base, RAG pipeline	Completed architecture + 250k-chunk corpus in Qdrant
M4–M6	Disease agent training, Nutrition fusion, ASR pipeline	Trained ViT-L+LoRA, Cross-modal MLP, Whisper Indic
M7–M9	Orchestrator, React PWA, Integration + testing	Full system integrated, Load tested, Latency validated
M10–M12	Evaluation, Expert panel, Thesis writing	MIRAGE/AgMMU results, Agronomist panel report, Thesis

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